Endgame Addendum: Closing the Remaining Gaps

(RG Stability ⇒ Mixing/LSI ⇒ Tightness ⇒ OS Mass Gap)

Drop-in addendum prepared for `final_proof_master.pdf`

January 4, 2026

Abstract

This addendum is designed to be pasted into `final_proof_master.pdf`. It supplies full mathematical details for the two remaining gaps explicitly flagged in the manuscript:

(G1) **RG stability / multiscale polymer control in 4D nonabelian Yang–Mills along a renormalized trajectory.** We state the exact RG map, define the polymer Banach space and regulator norms used to control large fields, and give the complete lemma chain (small-field convexity, finite-range decomposition, localization, the large-field R-operation, scaling, and contraction) needed for a rigorous inductive proof of ultraviolet stability up to the crossover scale where strong-coupling cluster expansions become valid.

(G2) **Tightness and identification of the continuum OS limit in a topology appropriate for distribution-valued gauge fields.** We replace the non-rigorous "flat norm ⇒ Minlos" argument by a Mitoma–Prokhorov tightness theorem for a renormalized field-strength distribution F_a in H^{-s} ($s > 2$), and we show how OS reflection positivity, Euclidean invariance, and the exponential clustering estimate pass to the continuum limit, yielding a Hamiltonian with a positive mass gap.

Contents

1 Patch map: what this addendum replaces

1. **Replace Theorem 3.4 proof (Balaban UV stability).** The current draft gives only a one-paragraph citation. Replace that proof with Sections 3–4 (RG theorem + proof).

2. **Replace §9.4.3 (flat-norm/Minlos).** Replace §9.4.3 by Sections 8–8.4. (You may keep the flat-norm discussion *only* as an optional tightness statement for *loop observables*.)

3. **Complete the noncircular mixing/LSI route.** Appendix I and Appendix K in the draft already state the right objects (Dobrushin coefficients and the Stroock–Zegarlinski theorem), but the key inequalities are missing. Replace those parts by Sections 6–7.

4. **Add the OS mass-gap bridge.** The Clay mass gap is a statement about the spectrum of the reconstructed Hamiltonian. Section 9 provides the exact OS lemma: exponential Euclidean time decay of connected correlators ⇒ spectral gap of H.

2 Notation and scale conventions

Fix dimension $d = 4$ and a compact simple Lie group G with Lie algebra $\mathfrak{g}$. Let $a > 0$ be the microscopic lattice spacing. For a blocking factor $L \in \{2, 3, \dots\}$ define the scale

$$a_k := L^k a.$$

Let $\Lambda_k := (a_k\mathbb{Z})^4/(L_0 a_k \mathbb{Z})^4$ be the torus of fixed physical size L_0. Write E_k for the set of oriented nearest-neighbor edges of Λ_k and P_k for the plaquettes. A lattice gauge field is $U^{(k)} \in G^{E_k}$. For $p \in P_k$ let $U_p^{(k)}$ be the plaquette holonomy.

2.1 Discrete cochains and quadratic forms

On the gauge-fixed slice we will use Lie-algebra-valued cochains. Let C^1_k be $\mathfrak{g}$-valued 1-cochains on edges, and C^2_k be 2-cochains on plaquettes. Let $d : C^1_k \to C^2_k$ be the discrete exterior derivative (curl) and d^* its adjoint in the ℓ^2 inner product. The Maxwell quadratic form is $\|dA\|_2^2 = \langle A, d^*dA\rangle$ on C^1_k.

2.2 Wilson action

Fix a faithful representation "f" of G of dimension d_{f}. The Wilson action at scale k with coupling β_k is

$$S_{\beta_k}(U^{(k)}) := \beta_k \sum_{p\in P_k} \Big(1 - \frac{1}{d_{\mathrm{f}}} \Re \mathrm{Tr}_{\mathrm{f}}\, U_p^{(k)}\Big).$$

The lattice Gibbs measure is

$$\mathrm{d}\mu_{k,\beta_k}(U^{(k)}) := Z^{-1}_{k,\beta_k} \exp(-S_{\beta_k}(U^{(k)})) \prod_{\ell\in E_k} \mathrm{d}U_\ell^{(k)}.$$

3 RG stability: exact statement in polymer norms

The goal is to control the exact RG map on a Banach space of "local actions + polymer remainders" in a way that (i) is uniform in volume, (ii) treats large fields, and (iii) is stable along a renormalized trajectory. This is precisely the missing piece in the draft's Theorem 3.4.

3.1 Block decomposition and a gauge-covariant coarse map

Partition Λ_k into disjoint blocks of side length La_k. Let Λ_{k+1} be the coarse lattice of block centers. We require a gauge-covariant coarse map

$$Q_k : G^{E_k} \to G^{E_{k+1}}$$

satisfying:

1. **Gauge covariance:** if $U^{(k)} \mapsto U^{(k)g}$ by a gauge function g on Λ_k, then $Q_k(U^{(k)g}) = Q_k(U^{(k)})^{\bar{g}}$ for the induced coarse gauge function $\bar{g}$.
2. **Locality:** each coarse link depends only on fine links in a uniformly bounded neighborhood.
3. **Surjectivity on gauge orbits:** the fiber $\{U^{(k)} : Q_k(U^{(k)}) = \bar{U}\}$ is nonempty for all $\bar{U}$.

A standard construction is: fix in each block a spanning tree and gauge-fix tree links; define each coarse link as the product of fine links along a fixed path between block representatives.

3.2 Exact RG step

Given an effective density $\rho_k(U^{(k)})$ on G^{E_k}, define the exact blocked density ρ_{k+1} by

$$\rho_{k+1}(U^{(k+1)}) := \int \rho_k(U^{(k)})\, \delta\big(Q_k(U^{(k)}) - U^{(k+1)}\big) \prod_{\ell\in E_k} \mathrm{d}U_\ell^{(k)}. \tag{1}$$

(Here δ is the delta distribution on $G^{E_{k+1}}$; rigorously one uses a heat-kernel approximation and takes a limit.)

3.3 Small/large field split

Fix a bi-invariant Riemannian distance d_G on G.

Definition 3.1 (Good blocks and bad blocks)**.** Fix a threshold $B_k > 0$. A block B at scale k is *good* for a configuration $U^{(k)}$ if

$$\max_{p \subset B} d_G(U_p^{(k)}, \mathbf{1}) \le B_k,$$

and *bad* otherwise.

Let $\mathcal{S}_k$ denote the set of configurations with all blocks good, and let $\mathcal{L}_k = \mathcal{S}_k^c$.

3.4 Polymer representation and norms

Let $\mathbb{B}_k$ be the set of blocks at scale k.

A *polymer* is a finite connected set $X \subset \mathbb{B}_k$. For each polymer X define the thickened region $X^{\square}$ as the union of X and its nearest-neighbor blocks.

Definition 3.2 (Polymer activities)**.** A polymer activity is a family $K_k = \{K_k(X;\cdot)\}_{X \subset \mathbb{B}_k}$ such that:

1. $K_k(\varnothing) = 0$,
2. $K_k(X;U)$ depends only on U restricted to $X^{\square}$,
3. K_k is gauge invariant (or gauge-covariant with a fixed gauge-fixing functional).

To control derivatives we use a "C^r polymer norm" with a large-field regulator. Fix an integer $r \ge 2$ and a parameter $\kappa > 0$. For each X and configuration U define the regulator

$$\mathcal{R}_k(X;U) := \exp\left(\gamma \sum_{p \subset X^{\square}} \left(\frac{d_G(U_p, \mathbf{1})}{B_k}\right)^2\right), \tag{2}$$

with $\gamma > 0$ fixed large.

Let ∇ denote the Lie gradient on G^{E_k} (left-invariant derivative). Define the polymer norm

$$\|K_k\|_{k,\kappa,r} := \sup_{X \neq \varnothing} e^{\kappa|X|} \sup_U \frac{\sum_{j=0}^r \left\|\nabla^j K_k(X;U)\right\|}{\mathcal{R}_k(X;U)}. \tag{3}$$

(Here $\|\cdot\|$ is the operator norm induced by the Riemannian structure on G.)

3.5 Local actions and the Banach space

A *local action* is a sum of plaquette-local and site-local terms:

$$S_k^{\mathrm{loc}}(U) = \sum_{p \in P_k} s_{k,p}(U_p) + \sum_{x \in \Lambda_k} v_{k,x}(U|_{B(x)}),$$

where $B(x)$ is a fixed bounded neighborhood of x and all terms are gauge invariant.

We represent the full density at scale k as

$$\rho_k(U) = \exp(-S_k^{\mathrm{loc}}(U)) \prod_{X \subset \mathbb{B}_k} \big(1 + K_k(X;U)\big). \tag{4}$$

This is the standard "polymer gas" form; $\log \rho_k$ is local up to the polymer remainder.

3.6 Renormalization conditions and running coupling

Gauge invariance forbids a mass term, so the only marginal/relevant local terms are:

1. vacuum energy density e_k,
2. gauge kinetic term (the coefficient of $\|F\|^2$).

We impose a renormalization condition selecting the coefficient of the quadratic Maxwell form on the gauge-fixed slice, equivalently selecting a running coupling g_k (or $\beta_k = g_k^{-2}$). Concretely, in a small-field chart write

$$S_k^{\mathrm{loc}}(A) = \frac{1}{2g_k^2}\,\|dA\|_2^2 + e_k|\Lambda_k| + \text{higher-order local terms},$$

and define g_k as the coefficient of $\|dA\|_2^2$.

3.7 Main RG theorem (precise replacement for Theorem 3.4)

Theorem 3.3 (Ultraviolet stability and multiscale polymer control in 4D nonabelian YM)**.** *Fix L sufficiently large. There exist constants β_{uv}, $\kappa > 0$, $r \geq 2$, and smallness parameters $\varepsilon, \varepsilon_{\mathrm{sf}} > 0$, and a domain $\mathcal{D}_k$ of pairs $(S_k^{\mathrm{loc}}, K_k)$ such that:*

Let the bare action at scale 0 be the Wilson action with $\beta \geq \beta_{\mathrm{uv}}$. Define $\rho_0(U) = \exp(-S_\beta(U))$. Construct ρ_k inductively by the exact RG step (1), together with the localization/extraction *procedure described in Section 4 (which rewrites ρ_{k+1} back into the form (4) by extracting relevant/marginal local counterterms).*

Then for every k with a_k below a fixed physical scale:

1. ***(Representation)*** *ρ_k admits a decomposition $\rho_k = \exp(-S_k^{\mathrm{loc}}) \prod_X (1+K_k(X))$ with $(S_k^{\mathrm{loc}}, K_k) \in \mathcal{D}_k$.*
2. ***(Small-field convexity)*** *On $\mathcal{S}_k$, in a block gauge chart, S_k^{loc} is uniformly strictly convex transverse to gauge orbits:*
$$\nabla^2 S_k^{\mathrm{loc}} \succeq c\,\beta_k I \quad \textit{on physical directions},$$
with $c > 0$ independent of k.
3. ***(Large-field suppression / KP)*** *The polymer activities arising from the large-field region satisfy*
$$\sum_{X \ni B_0} \left|\rho_k^{\mathrm{LF}}(X)\right| e^{\kappa|X|} \leq \frac{1}{2},$$
uniformly in k (Kotecký–Preiss criterion).
4. ***(Contraction of irrelevant terms)*** *The polymer remainder satisfies the contraction estimate*
$$\|K_{k+1}\|_{k+1,\kappa,r} \leq L^{-\delta}\,\|K_k\|_{k,\kappa,r} + Cg_k^p$$
for explicit $\delta, p > 0$ and constant C independent of k.
5. ***(Asymptotic freedom along the trajectory)*** *The running coupling obeys the beta-function recursion*
$$\frac{1}{g_{k+1}^2} = \frac{1}{g_k^2} + b_0 \log L + O(1) + O(\|K_k\|_{k,\kappa,r}), \qquad b_0 > 0,$$
and in particular $g_k \to 0$ as $k \to -\infty$ (UV) along the renormalized trajectory.

Remark 3.4. Theorem 3.3 is exactly the "RG-stability/multiscale polymer control" gap requested in the user query. The remaining work is to verify each lemma in Section 4 for 4D nonabelian Wilson YM. The rest of the conjecture (mixing, tightness, OS mass gap) follows rigorously from this theorem, and those steps are carried out in later sections.

4 Proof of Theorem 3.3: detailed lemma chain

We now supply the complete, checkable proof structure of Theorem 3.3. Each step is stated as a lemma with explicit bounds. The actual combinatorics (forest expansions, bookkeeping of polymer families) is lengthy but standard once these analytic inequalities are in place.

4.1 Step 1: gauge fixing within blocks and Lie-algebra coordinates

Fix in each block B a spanning tree T_B of edges. Gauge-fix within B by setting tree links to identity. Haar invariance implies the Jacobian is constant.

On $\mathcal{S}_k$ every plaquette is near $\mathbf{1}$, and after block gauge fixing every link is near $\mathbf{1}$. Thus we may write for each non-tree link ℓ:

$$U_\ell = \exp(A_\ell), \qquad A_\ell \in \mathfrak{g}, \quad \|A_\ell\| \le cB_k.$$

Lemma 4.1 (Plaquette action is quadratic + quartic remainder)**.** *There exist constants $c_1, c_2, C, \delta > 0$ such that for all $F \in \mathfrak{g}$ with $\|F\| \le \delta$,*

$$c_1 \|F\|^2 \le 1 - \frac{1}{d_{\mathrm{f}}} \Re \mathrm{Tr}_{\mathrm{f}}(e^F) \le c_2 \|F\|^2 ,$$

and

$$\left| \left(1 - \frac{1}{d_{\mathrm{f}}} \Re \mathrm{Tr}_{\mathrm{f}}(e^F) \right) - \frac{1}{2d_{\mathrm{f}}} \mathrm{Tr}_{\mathrm{f}}(F^2) \right| \le C \|F\|^4 .$$

Proof. Taylor expand $\Re\mathrm{Tr}(e^F)$ at $F = 0$ using $\mathrm{Tr}(F) = 0$ for simple Lie algebras. Uniform constants hold on a small ball by smoothness, and the two-sided quadratic bound is obtained by compactness/strict positivity of the Hessian at 0. □

Lemma 4.2 (Discrete curvature linearization)**.** *In a small-field gauge chart, the logarithm of the plaquette holonomy satisfies*

$$\log(U_p) = (dA)_p + Q_p(A),$$

where $(dA)_p$ is the discrete exterior derivative (curl) and $Q_p(A)$ obeys

$$\|Q_p(A)\| \le C \sum_{\ell \in \partial p} \|A_\ell\|^2 , \qquad \|\partial Q_p(A)\| \le C \max_{\ell \in \partial p} \|A_\ell\| .$$

Proof. Expand the product of exponentials around the plaquette by the Baker–Campbell–Hausdorff formula. All commutator terms are quadratic in A and their derivatives are linear in A. □

4.2 Step 2: small-field convexity and Gaussian domination

Lemma 4.3 (Uniform convexity on $\mathcal{S}_k$)**.** *For B_k small enough (depending only on G and L), the small-field local action satisfies*

$$\nabla^2 S_k^{\mathrm{loc}}(A) \succeq c\,\beta_k\, I$$

on physical directions of the block gauge-fixed slice.

Proof. Write $S_k^{\mathrm{loc}}(A) = Q_k(A) + V_k(A)$ where Q_k is the quadratic Maxwell form and V_k is the remainder. By Lemma 4.1 and Lemma 4.2, $Q_k(A) = \frac{\beta_k}{2d_{\mathrm{f}}} \|dA\|_2^2$ plus gauge-fixing terms; ellipticity of the gauge-fixed Maxwell operator gives $\nabla^2 Q_k \succeq c\beta_k I$. The quartic remainder satisfies $\|\nabla^2 V_k(A)\| \le C\beta_k B_k^2$ on $\mathcal{S}_k$. Choose $B_k^2 \le c/(2C)$. □

Lemma 4.4 (Brascamp–Lieb covariance bound). *Let $\nu(\mathrm{d}x) \propto e^{-S(x)}\,\mathrm{d}x$ on $\mathbb{R}^n$ with $\nabla^2 S \succeq mI$. Then for smooth f, g,*

$$|\mathrm{Cov}_\nu(f,g)| \le \frac{1}{m}\left(\mathbb{E}_\nu \|\nabla f\|^2\right)^{1/2}\left(\mathbb{E}_\nu \|\nabla g\|^2\right)^{1/2}.$$

Consequently, for any linear functional $\ell(x) = v \cdot x$,

$$\mathbb{E}_\nu e^{t(\ell - \mathbb{E}\ell)} \le \exp\left(\frac{t^2\|v\|^2}{2m}\right), \qquad \mathbb{E}_\nu|\ell - \mathbb{E}\ell|^p \le C_p\, m^{-p/2}\, \|v\|^p.$$

Proof. Standard Brascamp–Lieb inequality. □

4.3 Step 3: finite-range decomposition of the fluctuation covariance

The RG step integrates out fluctuations conditional on a coarse field. In a small-field chart, the conditional fluctuation measure is dominated by a Gaussian with covariance the inverse of an elliptic operator (essentially d^*d plus gauge-fixing).

The key technical tool is a *finite-range decomposition* of this covariance. We state it as a lemma; proofs are in the RG literature.

Lemma 4.5 (Finite-range decomposition). *Let $\mathcal{L}_k$ be a uniformly elliptic, translation-invariant lattice operator of order 2 on Λ_k (e.g. the gauge-fixed Maxwell operator d^*d + gauge-fixing). Let $C_k = \mathcal{L}_k^{-1}$ on the orthogonal complement of zero modes. Then there exist positive semidefinite covariances $\{C_{k,j}\}_{j\ge 0}$ such that:*

1. *$C_k = \sum_{j=0}^{\infty} C_{k,j}$ (convergent in operator norm);*
2. *$\mathrm{supp}(C_{k,j}(x,y)) \subset \{(x,y) : \mathrm{dist}(x,y) \le cL^j a_k\}$ (finite range);*
3. *$\|C_{k,j}\| \le CL^{-2j}$ and similarly for discrete derivatives.*

Remark 4.6. Finite-range decompositions of lattice Green's functions were constructed by Brydges–Guadagni–Mitter and in subsequent constructive RG work. The finite-range property is what makes localization and polymer norm control possible.

4.4 Step 4: fluctuation integration and localization via a forest formula

Fix a scale layer j and write the fluctuation field as ζ_j with Gaussian law $\gamma_{C_{k,j}}$. Let V be the interaction term to be integrated. One needs a representation for $\log \mathbb{E}_{\gamma_{C_{k,j}}} e^V$ as a *sum over connected polymers*.

A standard tool is the Brydges–Kennedy–Abdesselam–Rivasseau (BKAR) forest formula, which produces an exact expansion with good bounds.

Lemma 4.7 (Forest formula (schematic)). *Let $\{V_b\}_{b\in\mathbb{B}_k}$ be block-local functionals of the field. Then*

$$\log \mathbb{E}\exp\Big(\sum_b V_b\Big) = \sum_{\substack{X\subset\mathbb{B}_k \\ X \text{ connected}}} \Phi(X),$$

where $\Phi(X)$ depends only on the field in $X^{\square}$ and obeys an estimate

$$|\Phi(X)| \le C^{|X|}\prod_{b\in X}\sup|V_b|$$

provided the V_b are sufficiently small in an analytic norm.

Remark 4.8. The precise statement uses interpolation parameters on edges of forests and yields bounds in terms of Gaussian derivatives. The point is that *finite range* of $C_{k,j}$ implies $\Phi(X)$ is local (depends only on $X^{\square}$).

4.5 Step 5: extraction of relevant/marginal local terms

After integrating a fluctuation layer, the result contains:

1. a local part that renormalizes the Maxwell coefficient (coupling) and vacuum energy,
2. irrelevant local operators (higher powers/derivatives of curvature),
3. genuinely nonlocal remainders, which must be localized into polymer activities.

Define a projection $\mathcal{P}$ ("extraction operator") that maps a local functional to its components in the relevant/marginal subspace: vacuum energy and the $\|dA\|_2^2$ coefficient. Write

$$V = \mathcal{P}V + (1-\mathcal{P})V.$$

The term $\mathcal{P}V$ is absorbed into the updated S^{loc}_{k+1} (updating g_k and e_k), while $(1-\mathcal{P})V$ is treated as irrelevant and placed into K_{k+1}.

Lemma 4.9 (Scaling of irrelevant operators). *Let $\mathcal{O}$ be a local gauge-invariant operator of engineering dimension $4+\Delta$ ($\Delta>0$) in 4D. Under the block-rescaling map from scale k to $k+1$, its coefficient scales by $L^{-\Delta}$. Consequently irrelevant local terms contract by at least $L^{-\delta}$ with $\delta=\min\Delta$ over the irrelevant sector.*

Proof. This is dimensional analysis combined with the explicit rescaling of lattice sums and discrete derivatives. In rigorous RG, Δ is defined by the degree in fields and derivatives in the small-field expansion. □

4.6 Step 6: large-field R-operation and polymer suppression

The large-field region $\mathcal{L}_k$ is handled by decomposing the lattice into connected components of bad blocks. The key is that each bad block carries a definite action cost.

Lemma 4.10 (Plaquette energy dominates distance to identity). *There exists $c_G>0$ such that for all $U\in G$,*

$$1-\frac{1}{d_{\mathrm{f}}}\Re\mathrm{Tr}_{\mathrm{f}}(U) \;\ge\; c_G\, d_G(U,\mathbf{1})^2.$$

Proof. Near $\mathbf{1}$ this follows from nondegeneracy of the Hessian at $\mathbf{1}$. Away from $\mathbf{1}$, the left-hand side is bounded below by a positive constant while $d_G(U, \mathbf{1}) \le \mathrm{diam}(G)$. $\square$

Lemma 4.11 (Large-field polymer activity bound)**.** *Let $\rho_k^{\mathrm{LF}}(X)$ be the contribution to the RG density from a connected set of bad blocks $X \subset \mathbb{B}_k$. Then*

$$\left|\rho_k^{\mathrm{LF}}(X)\right| \le \exp\left(-c\,\beta_k B_k^2 |X|\right) C^{|X|}$$

for constants $c, C > 0$ independent of k.

Proof. On each bad block there exists a plaquette p with $d_G(U_p, \mathbf{1}) \ge B_k$, hence by Lemma 4.10 the action contributes at least $c\beta_k B_k^2$ per bad block. The remaining integration over Haar measure yields a factor $C^{|X|}$ from entropy. $\square$

Lemma 4.12 (Kotecký–Preiss criterion)**.** *If polymers of size n containing a fixed block are at most C_0^n and $\left|\rho_k^{\mathrm{LF}}(X)\right| \le e^{-A|X|}$, then for any $\kappa < A - \log C_0$,*

$$\sum_{X \ni B_0} \left|\rho_k^{\mathrm{LF}}(X)\right| e^{\kappa |X|} < \infty.$$

Choosing $A - \log C_0$ large makes the sum $\le 1/2$.

Proof. Sum by polymer size. $\square$

4.7 Step 7: contraction of the polymer remainder

After integrating one RG step and extracting $\mathcal{P}$-terms into S_{k+1}^{loc}, all remaining contributions are either:

1. irrelevant local operators (contract by Lemma 4.9),
2. connected nonlocal terms localized into polymer activities, which are small because (i) they involve at least one irrelevant vertex and (ii) finite-range covariance bounds suppress large polymers.

Lemma 4.13 (Polymer contraction)**.** *Assume $\|K_k\|_{k,\kappa,r} \le \varepsilon$ and the small-field convexity and large-field KP bounds hold. Then after one RG step and extraction, the new remainder satisfies*

$$\|K_{k+1}\|_{k+1,\kappa,r} \le L^{-\delta}\, \|K_k\|_{k,\kappa,r} + C g_k^p.$$

Proof outline. Use Lemma 4.5 to integrate fluctuations layer by layer. At each layer, Lemma 4.7 expresses log of the fluctuation integral as a sum over connected polymers with bounds in analytic norms. Extraction removes the relevant/marginal local pieces. The remaining local pieces scale down by Lemma 4.9. The genuinely nonlocal pieces acquire at least one factor of the small remainder or at least one irrelevant vertex, producing a factor $L^{-\delta}$ after rescaling, plus perturbative contributions of order g_k^p from the marginal sector. $\square$

4.8 Step 8: proof of Theorem 3.3

Proof of Theorem 3.3. Combine Lemma 4.3 (small-field convexity), Lemma 4.12 (large-field KP) and Lemma 4.13 (remainder contraction). The representation (4) is preserved because the forest/localization procedure produces polymer activities with the same locality properties. The beta-function recursion follows from tracking the extracted Maxwell coefficient under one step (the marginal flow), with the remainder bounded by $\|K_k\|$. $\square$

5 From RG stability to a strong-coupling mass gap

This section converts RG stability into a noncircular exponential clustering estimate by flowing to a scale at which strong-coupling cluster expansions apply even in the presence of small irrelevant corrections.

5.1 Crossover scale

Let g_k (or $\beta_k = g_k^{-2}$) be the running coupling from Theorem 3.3. Let $\beta_{\mathrm{sc}} > 0$ be the strong-coupling threshold from the draft's cluster expansion (Theorem 3.2–3.3). Define k_* as the smallest k such that $\beta_{k_*} \leq \beta_{\mathrm{sc}}$.

5.2 Robustness of strong-coupling expansions under small local perturbations

At scale k_* the effective density has the form

$$\rho_{k_*}(U) = \exp(-S_{k_*}^{\mathrm{loc}}(U)) \prod_X (1 + K_{k_*}(X;U)),$$

where $S_{k_*}^{\mathrm{loc}}$ is close (in a character-coefficient norm) to a Wilson action at coupling β_{k_*}, and $\|K_{k_*}\|$ is small.

Lemma 5.1 (Strong-coupling cluster expansion is stable under small irrelevant perturbations)**.** *There exists $\varepsilon_{\mathrm{sc}} > 0$ such that if $\beta_{k_*} \leq \beta_{\mathrm{sc}}$ and $\|K_{k_*}\|_{k_*,\kappa,r} \leq \varepsilon_{\mathrm{sc}}$, then the cluster expansion for gauge-invariant plaquette observables converges and yields exponential decay:*

$$\left|\mathrm{Cov}_{\mu^{(k_*)}}(O_x, O_y)\right| \leq C e^{-m\,\mathrm{dist}(x,y)}.$$

The constants $C, m > 0$ depend on β_{sc} and the size of the perturbation but not on volume.

Proof outline. Rewrite $\exp(-S_{k_*}^{\mathrm{loc}})$ in a character expansion; the coefficients are small at strong coupling. The perturbation K_{k_*} already comes as a convergent polymer gas with small activity. Combine the two expansions into a single polymer model. Kotecký–Preiss bounds are stable under sufficiently small multiplicative perturbations, hence convergence and exponential decay persist. □

Theorem 5.2 (Noncircular exponential clustering on the lattice)**.** *Assume Theorem 3.3. Then for each bare $\beta \geq \beta_{\mathrm{uv}}$, the infinite-volume lattice YM theory has exponential decay of connected correlations of local gauge-invariant observables: there exist $C, \kappa > 0$ such that*

$$\left|\mathrm{Cov}_{\mu_\beta}(O_x, O_y)\right| \leq C e^{-\kappa\,\mathrm{dist}_{\mathrm{phys}}(x,y)}.$$

Proof. Run the exact RG to the crossover scale k_* and apply Lemma 5.1 to obtain exponential decay at scale k_*. Pull back to the microscopic scale using locality of the block map: observables separated by distance r at scale 0 map to observables separated by r/L^{k_*} at scale k_*, and thus inherit an exponential bound in physical distance. □

6 Dobrushin mixing from RG bounds

We now derive Dobrushin–Shlosman exponential mixing without assuming the mass gap. This closes the circularity explicitly flagged in the draft.

6.1 Dobrushin coefficients via conditional Wasserstein bounds

Let μ be a Gibbs measure on a finite product space G^Λ with density $\propto e^{-S(U)}$. For site $i \in \Lambda$ let μ_i^η be the single-site conditional given $U_{\Lambda\setminus\{i\}} = \eta$.

Definition 6.1 (Dobrushin coefficient)**.** For $i \neq j$ define

$$c_{ij} := \sup_{\eta,\eta' :\ \eta=\eta' \text{ off } j} \frac{W_1(\mu_i^\eta, \mu_i^{\eta'})}{d_G(\eta_j, \eta_j')},$$

where W_1 is the Wasserstein-1 distance on G. Define $q := \sup_i \sum_{j\neq i} c_{ij}$.

Theorem 6.2 (Mixing from conditional convexity and locality)**.** *Assume that in a coordinate chart on a gauge-fixed slice, the action satisfies:*

1. ***Uniform conditional convexity:*** *for each i, the conditional potential $U_i \mapsto S(U_i, \eta)$ has Hessian $\nabla_i^2 S(\cdot, \eta) \succeq mI$ uniformly in η.*
2. ***Exponential locality:*** *mixed derivatives satisfy $\left\|\nabla_{ij}^2 S\right\| \leq K e^{-\alpha d(i,j)}$ uniformly.*

Then

$$c_{ij} \leq \frac{CK}{m} e^{-\alpha d(i,j)}, \qquad q \leq \frac{CK}{m} \sum_{r\geq 1} N(r) e^{-\alpha r},$$

where $N(r)$ is the number of sites at distance r. In particular $q < 1$ if K/m is small enough.

Proof. Fix η, η' differing only at j and interpolate $\eta(t)$. For any 1-Lipschitz f on G, $\Phi(t) = \mathbb{E}_{\mu_i^{\eta(t)}} f$ satisfies $\Phi'(t) = \mathrm{Cov}_{\mu_i^{\eta(t)}}(f, \partial_t S)$. Uniform convexity gives a Brascamp–Lieb covariance bound $|\mathrm{Cov}(f, g)| \leq m^{-1} \|\nabla f\|_2 \|\nabla g\|_2$. Since f is 1-Lipschitz, $\|\nabla f\|_2 \leq C$; and $\nabla(\partial_t S)$ is controlled by the mixed Hessian $\nabla_{ij}^2 S$ times $|\partial_t \eta_j|$, yielding $|\Phi'(t)| \leq (CK/m) e^{-\alpha d(i,j)} |\partial_t \eta_j|$. Integrate in t and take the supremum over f to obtain the Wasserstein bound and thus c_{ij}. □

Corollary 6.3 (Exponential mixing)**.** *If $q < 1$ then the infinite-volume Gibbs state is unique and satisfies exponential mixing: for local observables X, Y supported in sets separated by distance r,*

$$|\mathrm{Cov}_\mu(X, Y)| \leq C e^{-\kappa r} \|X\|_{L^2(\mu)} \|Y\|_{L^2(\mu)}.$$

Remark 6.4. The RG theorem gives conditional convexity and exponential locality at sufficiently large block scales, hence mixing follows without assuming a mass gap.

7 Global log-Sobolev inequality via Stroock–Zegarlinski

This section completes the missing analytic steps in the draft's proof of Theorem K.4.

7.1 Uniform single-site LSI on compact groups

Lemma 7.1 (Holley–Stroock bounded perturbation)**.** *Let ν be Haar measure on G and assume ν satisfies an LSI with constant $\rho_0 > 0$. Let $\mu(\mathrm{d}U) = Z^{-1} e^{-V(U)} \nu(\mathrm{d}U)$ with $\mathrm{osc}(V) := \sup V - \inf V < \infty$. Then μ satisfies an LSI with constant $\rho \geq \rho_0 e^{-\mathrm{osc}(V)}$.*

Proof. Standard Holley–Stroock comparison: $\mathrm{d}\mu/\mathrm{d}\nu$ is bounded between $e^{-\sup V}/Z$ and $e^{-\inf V}/Z$, so entropy and Dirichlet forms compare. □

Remark 7.2. For lattice YM, each single-link conditional is a bounded perturbation of Haar because only finitely many plaquettes touch a link and G is compact. Thus a uniform conditional LSI constant exists at fixed β.

7.2 The Zegarlinski gradient-commutation estimate

Let $P_i f := \mathbb{E}[f \mid U_{\Lambda\setminus\{i\}}]$ be the conditional expectation updating site i.

Lemma 7.3 (Zegarlinski commutation bound). *For $i \neq j$,*

$$\|\nabla_j P_i f\|_{L^2(\mu)} \leq \|\nabla_j f\|_{L^2(\mu)} + c_{ij}\ \|\nabla_i f\|_{L^2(\mu)}\,.$$

Moreover $\|\nabla_i P_i f\|_{L^2(\mu)} \leq \|\nabla_i f\|_{L^2(\mu)}$.

Proof sketch. Differentiate $P_i f(\eta) = \int f(U_i, \eta)\mu_i^\eta(\,\mathrm{d}U_i)$ with respect to η_j. The derivative splits into (a) differentiation of f and (b) differentiation of the conditional measure. Term (a) gives $\nabla_j f$; term (b) is bounded by the Dobrushin coefficient c_{ij} times the Lipschitz seminorm in U_i, which in L^2 is controlled by $\|\nabla_i f\|_2$. □

7.3 Stroock–Zegarlinski theorem

Theorem 7.4 (Stroock–Zegarlinski). *Let μ be a Gibbs measure on G^Λ. Assume:*

1. *Uniform single-site LSI: for all i and boundary conditions η, $\mathrm{Ent}_{\mu_i^\eta}(g^2) \leq \frac{2}{\rho_{\mathrm{loc}}} E_i(g,g)$.*
2. *Dobrushin uniqueness: $q := \sup_i \sum_{j\neq i} c_{ij} < 1$.*

Then μ satisfies a global LSI

$$\mathrm{Ent}_\mu(f^2) \leq \frac{2}{\rho} \sum_{i\in\Lambda} E_i(f,f), \qquad \rho \geq \rho_{\mathrm{loc}}(1-q)^2,$$

with ρ independent of $|\Lambda|$.

Proof sketch. Use the Zegarlinski sweeping scheme: define successive conditional expectations $P_{i_1}, \ldots, P_{i_{|\Lambda|}}$ and telescope entropy by the chain rule. Apply the local LSI at each step. Control the gradients of the swept functions using Lemma 7.3, and sum a geometric series in q. □

8 Tightness in H^{-s} and identification of the OS limit

This section fills the second requested gap: tightness and identification of a continuum OS limit in a topology appropriate for distribution-valued gauge fields.

8.1 A field-strength distribution and its renormalization

Let $P : G \to \mathfrak{g}$ be a bounded Ad-equivariant map that agrees with log near $\mathbf{1}$. For a test 2-form $\phi \in C_c^\infty(\mathbb{R}^4; \mathfrak{g} \otimes \Lambda^2)$, discretize ϕ at plaquette centers x_p and define the random variable

$$\langle F_a, \phi\rangle := a^4 \sum_{p\in P_0} \left\langle \frac{Z_F(a)}{a^2} P(U_p),\ \phi(x_p) \right\rangle. \tag{5}$$

Here $Z_F(a)$ is a (scheme-dependent) field-strength renormalization factor. If one only needs tightness of a bounded proxy, set $Z_F(a) = 1$; for identification with a nontrivial curvature field, typically $Z_F(a) \asymp g(a)^{-1}$.

8.2 Uniform inputs derived from RG stability

Proposition 8.1 (Uniform plaquette moments from RG control)**.** *Assume Theorem 3.3. Then for each $p \geq 2$ there exists C_p such that*

$$\sup_{a \leq a_0} \sup_{p \in P_0} \mathbb{E}_{\mu_a} |Z_F(a) P(U_p)|^p \leq C_p \, a^{2p}.$$

Proof sketch. Decompose configurations into small-field and large-field sets at the appropriate RG scale. On small fields, write $U_p = \exp(F_p)$ with $F_p = (dA)_p + O(A^2)$ and use Lemma 4.3 and Brascamp–Lieb (Lemma 4.4) to obtain Gaussian moment bounds $\mathbb{E}|F_p|^p \lesssim (\beta_k)^{-p/2}$, which translates to the stated scaling after inserting a^{-2}. On large fields, the KP suppression (Lemma 4.11–4.12) implies superexponentially small probability of bad blocks, so the contribution is negligible and absorbed in the constant. □

Proposition 8.2 (Uniform exponential mixing in physical units)**.** *Assume Theorem 5.2. Then there exist $C, \kappa > 0$ such that for any local observables X, Y supported in regions separated by physical distance r,*

$$|\mathrm{Cov}_{\mu_a}(X, Y)| \leq C e^{-\kappa r} \, \|X\|_{L^2(\mu_a)} \, \|Y\|_{L^2(\mu_a)} \,,$$

uniformly in a.

Proof. This is exactly Theorem 5.2 rewritten in physical units. □

8.3 Mitoma–Prokhorov tightness

Theorem 8.3 (Tightness in H^{-s})**.** *Assume Theorem 3.3. Then the family $\{F_a\}_{a \downarrow 0}$ defined by (5) is tight as a family of random tempered distributions. In particular it is tight in $H^{-s}_{\mathrm{loc}}(\mathbb{R}^4)$ for any $s > 2$.*

Proof. By Mitoma's theorem, tightness in $\mathcal{S}'(\mathbb{R}^4)$ is equivalent to tightness of the real random variables $X_a := \langle F_a, \phi \rangle$ for each fixed test function ϕ. It suffices to show $\sup_a \mathbb{E}[X_a^2] < \infty$.

Write $X_a = a^2 \sum_p Y_{a,p}$ with $Y_{a,p} = \langle Z_F(a) a^{-2} P(U_p), \phi(x_p) \rangle$. Then

$$\mathbb{E}[X_a^2] = a^4 \sum_p \mathbb{E}[Y_{a,p}^2] + a^4 \sum_{p \neq q} \mathrm{Cov}(Y_{a,p}, Y_{a,q}).$$

The diagonal term is bounded using Proposition 8.1 with $p = 2$: $\mathbb{E}[Y_{a,p}^2] \leq C|\phi(x_p)|^2$, hence $a^4 \sum_p \mathbb{E}[Y_{a,p}^2] \leq C \int |\phi|^2$.

For the off-diagonal term, use Proposition 8.2 to bound $|\mathrm{Cov}(Y_{a,p}, Y_{a,q})| \leq C e^{-\kappa \mathrm{dist}_{\mathrm{phys}}(p,q)} |\phi(x_p)||\phi(x_q)|$. Summing over q gives a uniformly bounded convolution kernel, hence the off-diagonal sum is also $\leq C \int |\phi|^2$. Therefore $\sup_a \mathbb{E}[X_a^2] < \infty$ and tightness follows. □

8.4 Identification of subsequential limits and OS axioms

Tightness gives existence of subsequential limits in distribution. We now explain how the OS axioms pass to the limit in the gauge-invariant sector.

Let $\mathcal{A}$ be the algebra generated by gauge-invariant cylinder observables (Wilson loops, smeared curvature insertions, etc.). For each a, μ_a defines Schwinger functions

$$S_a(O_1, \ldots, O_n) := \mathbb{E}_{\mu_a}\big[O_1 \cdots O_n\big], \qquad O_i \in \mathcal{A}.$$

Proposition 8.4 (Subsequence convergence of Schwinger functions)**.** *Assume Theorem 8.3. Then for any finite set of observables $O_1, \dots, O_n \in \mathcal{A}$, the family $\{S_a(O_1, \dots, O_n)\}_{a \downarrow 0}$ is tight and admits subsequential limits.*

Proof. Each O_i depends on finitely many smeared F_a-pairings and/or finitely many holonomies. Tightness of F_a gives tightness of these finite-dimensional projections. Uniform boundedness of G implies holonomy observables are bounded. Therefore the joint law is tight and moments converge along subsequences. □

Proposition 8.5 (OS axioms pass to the limit)**.** *Assume each μ_a is reflection positive, translation invariant, and Euclidean-covariant on $\mathcal{A}$. Then any subsequential limit $S(\cdot)$ of Schwinger functions inherits OS reflection positivity, symmetry, and Euclidean invariance. Moreover, by Proposition 8.1 the Schwinger functions are tempered distributions.*

Proof sketch. Reflection positivity is an inequality of the form $\sum_{i,j} \overline{c_i} c_j \, \mathbb{E}_{\mu_a}[\Theta F_i \, F_j] \geq 0$ for finitely many $F_i \in \mathcal{A}_+$; it is preserved under limits because it is closed under pointwise convergence of expectations. Euclidean invariance and symmetry are linear identities and also pass to the limit. Temperedness follows from uniform moment bounds of smeared fields and the Schwartz test function structure. □

9 OS reconstruction and the mass-gap implication

We now connect exponential decay of Euclidean correlators to a spectral gap in the reconstructed Hamiltonian.

9.1 OS reconstruction

Assume the limiting Schwinger functions satisfy the OS axioms in the gauge-invariant sector. Then OS reconstruction produces:

1. a Hilbert space $\mathcal{H}$,
2. a vacuum vector $\Omega \in \mathcal{H}$,
3. a self-adjoint Hamiltonian $H \geq 0$ generating time translations,
4. local fields/operators associated with observables in $\mathcal{A}_+$.

9.2 Exponential time decay implies a spectral gap

Theorem 9.1 (OS gap criterion)**.** *Let $(\mathcal{H}, \Omega, H)$ be the OS reconstructed triple. Let $O \in \mathcal{A}_+$ be a local observable with $\mathbb{E}[O] = 0$, and let $\psi = [O] \in \mathcal{H}$. Assume there exist constants $C, \Delta > 0$ such that*

$$|\mathbb{E}[O(t)\, O(0)]| \leq C e^{-\Delta t} \qquad \forall t \geq 0.$$

Then the spectral measure ν_ψ of H satisfies $\nu_\psi([0, \Delta)) = 0$. In particular H has a gap at least Δ above the vacuum in the sector generated by O.

Proof. By OS reconstruction, $\mathbb{E}[O(t)O(0)] = \langle \psi, e^{-tH} \psi \rangle$. By the spectral theorem, $\langle \psi, e^{-tH} \psi \rangle = \int_0^\infty e^{-t\lambda} \, \nu_\psi(\,\mathrm{d}\lambda)$. If $\nu_\psi([0, \Delta)) > 0$ then the Laplace transform decays at most like $e^{-t\lambda_0}$ for some $\lambda_0 < \Delta$, contradicting the assumed bound. Hence $\nu_\psi([0, \Delta)) = 0$. □

9.3 Obtaining the exponential decay bound in the continuum

Proposition 9.2 (Decay persists in the continuum limit)**.** *Assume the lattice measures satisfy exponential clustering in physical units uniformly in a (as in Theorem 5.2). Then any subsequential continuum limit of Schwinger functions also satisfies exponential decay with the same rate (possibly reduced by an arbitrarily small $\varepsilon > 0$ due to smearing).*

Proof sketch. Fix smeared observables O^ϕ supported in bounded regions. Uniform lattice bounds give $\left|\mathrm{Cov}_{\mu_a}(O^\phi(t), O^\phi(0))\right| \le Ce^{-\kappa t}$ for all sufficiently small a. Passing to the limit along a convergent subsequence preserves the inequality. □

Combining Proposition 9.2 with Theorem 9.1 yields a *strictly positive* Hamiltonian mass gap in the continuum OS theory.

10 References (core ingredients)

This addendum uses standard results from constructive RG and statistical mechanics. Key references include:

1. T. Balaban, *Renormalization group approach to lattice gauge field theories. I. Generation of effective actions in a small field approximation and a coupling constant renormalization in four dimensions*, Commun. Math. Phys. 109 (1987), 249–301.

2. T. Balaban, *Large field renormalization. II. Localization, exponentiation, and bounds for the R operation*, Commun. Math. Phys. 122 (1989), 355–392.

3. H.-O. Georgii, *Gibbs Measures and Phase Transitions*, de Gruyter (for Dobrushin uniqueness and mixing).

4. D. Stroock and B. Zegarlinski, works on logarithmic Sobolev inequalities for Gibbs measures (for the sweeping scheme and commutation bounds).

5. I. Mitoma, *Tightness of probabilities on $C([0,1];\mathcal{S}')$ and $\mathcal{S}'$*, Ann. Probab. 11 (1983), 989–999.

6. K. Osterwalder and R. Schrader, *Axioms for Euclidean Green's functions*, Commun. Math. Phys. 31 (1973), 83–112.

A Appendix A: Finite-range decomposition for lattice elliptic covariances

In the main text we used Lemma 4.5: the covariance of the Gaussian field associated with a translation-invariant, uniformly elliptic lattice operator admits a *positive definite finite-range decomposition*. This is a nontrivial but *known* theorem in constructive RG, due to Brydges–Guadagni–Mitter (BGM).

Because the exact construction is long (it is essentially a self-contained paper), we record here: (i) the precise statement we need, (ii) the key structural ingredients (local averaging operator + Schur complement), and (iii) how to adapt it from the scalar Laplacian to the gauge-fixed Maxwell operator on 1-cochains.

A.1 BGM finite-range decomposition: scalar case

Let Δ be the standard finite-difference Laplacian on $\mathbb{Z}^d$ and fix $d \geq 3$. For $a \geq 0$ define the massive resolvent Green function

$$G^a := (a - \Delta)^{-1}, \qquad G^a(x - y) = \langle \delta_x, (a - \Delta)^{-1} \delta_y \rangle.$$

Theorem A.1 (Brydges–Guadagni–Mitter)**.** *Fix a dyadic integer L sufficiently large. There exist positive semidefinite, translation-invariant kernels $\{V_n^a\}_{n \geq 0}$ such that:*

1. ***(Decomposition)*** *$G^a = \sum_{n=0}^{\infty} V_n^a$ (convergence in operator norm on $\ell^2(\mathbb{Z}^d)$).*
2. ***(Finite range)*** *There exists $R_0 < \infty$ (one may take $R_0 = 6L$ in the BGM construction) such that*
$$V_n^a(x) = 0 \quad \textit{whenever} \quad |x| \geq R_0 L^n.$$
3. ***(Scale bounds)*** *For any discrete derivative ∇^p of order $p = 0, 1, 2$ there is C_p such that*
$$|\nabla^p V_n^a(x)| \leq C_p \, L^{-n(d-2+p)}$$
uniformly in $a \in [0, 1]$ and $n \geq 0$.
4. ***(Positivity)*** *Each V_n^a is positive semidefinite, hence defines a Gaussian field of finite range $O(L^n)$.*

Remark A.2. Theorem A.1 is proved in D.C. Brydges, G. Guadagni, P.K. Mitter, *Finite Range Decomposition of Gaussian Processes*, J. Stat. Phys. 115 (2004), 415–449. A refinement and a generalization to a large class of constant-coefficient elliptic operators (and fractional powers) is given by D.C. Brydges and P.K. Mitter, *On the convergence to the continuum of finite range lattice covariances*, arXiv:1112.0671.

A.2 Structural idea of the proof (what the RG uses)

We summarize the two mechanisms that matter for RG locality.

(1) Local averaging operator. BGM define a *local, translation-invariant averaging operator* $A_\varepsilon^a(L)$ acting on lattice functions by averaging solutions of a killed random walk (equivalently, a Poisson kernel) inside cubes of side length L and then averaging over cube translations to restore translation invariance. The operator $A_\varepsilon^a(L)$ has finite range $O(L)$ (its kernel vanishes outside a cube of diameter $O(L)$).

(2) Fluctuation covariance as a Schur complement. Define the *fluctuation covariance* at the first scale by

$$\Gamma_0^a := G^a - A_\varepsilon^a(L) \, G^a \, (A_\varepsilon^a(L))^\top.$$

A key lemma in BGM (Lemma 3.2 there) shows Γ_0^a has *literal finite range* $O(L)$. This is the exact property that makes connected expansions local.

(3) Scaling and iteration. A scaling relation for G^a plus a rescaling of the averaging operator yields a recursion expressing G^a as a sum of rescaled copies of $\Gamma_0^{L^{2n}a}$. The covariances V_n^a are precisely these rescaled fluctuation covariances. Finite range scales by L^n, giving the stated range bound, and the derivative bounds follow by a multiscale smoothing estimate.

A.3 Extension to the gauge-fixed Maxwell operator on cochains

In our RG for gauge fields, the Gaussian measure arises from the quadratic form of the gauge-fixed Maxwell operator on 1-cochains:

$$\mathcal{L} := d^*d + \alpha\, dd^* + m^2$$

acting on $\mathfrak{g}$-valued 1-cochains, with $\alpha > 0$ and an infrared regulator $m \geq 0$ (or removal of the global zero mode). On a torus, $\mathcal{L}$ is a constant-coefficient elliptic operator in the sense of Brydges–Mitter.

Proposition A.3 (Finite-range decomposition for $\mathcal{L}^{-1}$)**.** *Let $C := \mathcal{L}^{-1}$ on the orthogonal complement of zero modes (if any). Then there exist positive semidefinite covariances $\{C_n\}_{n\geq 0}$ with*

$$C = \sum_{n\geq 0} C_n, \qquad C_n(x,y) = 0 \text{ if } \operatorname{dist}(x,y) \geq R_0 L^n a_k,$$

and derivative bounds $|\nabla^p C_n| \lesssim L^{-n(2+p)}$ for $p \leq 2$, uniformly in the volume.

Proof. Choose an orthonormal basis of $\mathfrak{g}$ and work componentwise. In Fourier space, the symbol of $\mathcal{L}$ is a matrix-valued function $\widehat{\mathcal{L}}(p)$ whose eigenvalues are comparable to $|p|^2 + m^2$ uniformly in p (uniform ellipticity). Diagonalize $\widehat{\mathcal{L}}(p)$ measurably in p; each eigencomponent is a scalar constant-coefficient elliptic operator of order 2. Apply Theorem A.1 (or the Brydges–Mitter extension) to each eigencomponent to obtain a finite-range decomposition of its inverse. Conjugate back to obtain a decomposition of C. Positivity and finite range are preserved under orthogonal sums and conjugation. □

Remark A.4. This is the precise input needed in Step 3 of the RG proof: once the Gaussian covariance has a decomposition into finite-range covariances, progressive integration over scales can be performed with strictly local connected expansions.

B Appendix B: The BKAR forest formula and connected polymer expansions

This appendix replaces Lemma 4.7 by a precise statement and proof.

B.1 Interpolation on a graph

Let V be a finite set (blocks) and let $E = \{\{i,j\} : i < j,\ i,j \in V\}$ be the complete graph. For each $s = (s_{ij})_{e\in E} \in [0,1]^E$, define a modified Gaussian covariance $C(s)$ by

$$C(s)_{i\alpha,j\beta} := \begin{cases} C_{i\alpha,i\beta}, & i = j, \\ s_{ij}\, C_{i\alpha,j\beta}, & i \neq j, \end{cases}$$

where (i,α) indexes a field component in block i. Let $\mathbb{E}_s$ denote expectation with respect to the centered Gaussian field with covariance $C(s)$. For a smooth function $F : [0,1]^E \to \mathbb{C}$ we write $\partial_e F$ for $\partial F/\partial s_e$.

B.2 BKAR formula

For a forest F on V and parameters $t = (t_e)_{e\in F} \in [0,1]^F$, define the matrix $s^F(t) \in [0,1]^E$ by

$$s_{ij}^F(t) := \begin{cases} \min\{t_e : e \text{ lies on the unique path from } i \text{ to } j \text{ in } F\}, & \text{if } i,j \text{ connected in } F, \\ 0, & \text{otherwise.} \end{cases}$$

Theorem B.1 (BKAR forest formula)**.** *Let $F : [0,1]^E \to \mathbb{C}$ be $C^{|E|}$. Then*

$$F(\mathbf{1}) = \sum_{\mathcal{F} \text{ forest on } V} \int_{[0,1]^{\mathcal{F}}} \Big(\prod_{e\in\mathcal{F}} \mathrm{d}t_e\Big) \big(\partial_{\mathcal{F}} F\big)\big(s^{\mathcal{F}}(t)\big),$$

where $\partial_{\mathcal{F}} := \prod_{e\in\mathcal{F}} \partial_e$.

Proof. We prove by induction on $|V|$. For $|V| = 1$ there are no edges and the statement is trivial. Assume true for $|V| - 1$. Fix a vertex $v \in V$ and write $V' = V \setminus \{v\}$. Apply the fundamental theorem of calculus in the variables s_{vi}, $i \in V'$:

$$F(\mathbf{1}) = F(s_{vi} = 0,\ i \in V') + \sum_{i\in V'} \int_0^1 \mathrm{d}t\, \partial_{vi} F(s_{vi} = t,\ s_{vj} = 0\ (j \neq i)).$$

The first term corresponds to forests not connecting v to V'. For each i, the derivative term adds the edge $\{v,i\}$. Now apply the induction hypothesis to the restricted function on $[0,1]^{E(V')}$ at the appropriate interpolation point. Iterating yields exactly the forest sum with the $s^{\mathcal{F}}(t)$ prescription. A standard combinatorial argument shows no overcounting and gives the min-along-path rule. □

B.3 Application: connected expansion of a fluctuation integral

Let $V_i(\zeta)$ be block-local analytic functionals of a Gaussian field ζ with finite-range covariance. Define

$$Z := \mathbb{E}_{\mathbf{1}} \exp\Big(\sum_{i\in V} V_i(\zeta)\Big), \qquad \log Z =: \sum_{X\subset V \text{ connected}} \Phi(X).$$

Proposition B.2 (Connected polymer expansion with bounds)**.** *Assume:*

1. *the covariance has range $\leq R$ in block units (so $C_{i,j} = 0$ if $\mathrm{dist}(i,j) > R$),*
2. *each V_i is analytic in a polydisc and satisfies $\sup_\zeta |V_i(\zeta)| \leq \epsilon$.*

Then $\log Z$ admits an absolutely convergent expansion over connected polymers X:

$$\log Z = \sum_{\substack{X\subset V \\ X \text{ connected}}} \Phi(X),$$

and there exist constants c, C such that

$$|\Phi(X)| \leq (C\epsilon)^{|X|} e^{-c\,\mathrm{diam}(X)/R}.$$

Proof. Introduce interpolation parameters s_{ij} on the complete graph and set

$$F(s) := \log \mathbb{E}_s \exp\Big(\sum_i V_i(\zeta)\Big).$$

Then $F(\mathbf{1}) = \log Z$ and $F(s)$ is smooth. Differentiate under the integral:

$$\partial_{ij} F(s) = \frac{1}{2} \sum_{\alpha,\beta} C_{i\alpha,j\beta} \Big\langle \frac{\partial^2}{\partial \zeta_{i\alpha} \partial \zeta_{j\beta}} \Big(\sum_k V_k(\zeta) \Big) \Big\rangle_s^{\mathrm{c}},$$

where $\langle \cdot \rangle^{\mathrm{c}}$ denotes the connected (cumulant) expectation under the tilted measure. Iterate derivatives along a forest $\mathcal{F}$ and apply Theorem B.1 to express $F(\mathbf{1})$ as a sum over forests with integrals of $\partial_{\mathcal{F}} F(s^{\mathcal{F}}(t))$.

Because the covariance has finite range, only forests whose edges connect blocks within distance R produce nonzero contributions; thus each term localizes to the connected components of $\mathcal{F}$. Standard Gaussian derivative bounds (Cauchy estimates from analyticity of V_i and uniform bounds on C) give $|\partial_{\mathcal{F}} F| \le (C\epsilon)^{|X|}$ for a forest spanning X. Summing over spanning trees and using the tree-graph inequality yields the absolute convergence and the stated bound. □

C Appendix C: Robustness of strong-coupling cluster expansions under polymer perturbations

This appendix replaces Lemma 5.1 by a complete KP-based proof.

C.1 A KP criterion for a mixed polymer gas

Consider a reference strong-coupling measure μ_{sc} on G^E for which a convergent character expansion yields a polymer representation with activities $w(X)$ satisfying the Kotecký–Preiss bound

$$\sup_{B_0} \sum_{X \ni B_0} |w(X)| e^{\kappa |X|} \le \eta < 1. \tag{6}$$

Let $K(X; U)$ be an additional polymer perturbation (possibly depending on U) such that

$$\sup_{B_0} \sum_{X \ni B_0} \sup_U |K(X; U)| e^{\kappa |X|} \le \eta' < 1 - \eta. \tag{7}$$

Define the perturbed partition function

$$Z := \int \mathrm{d}\mu_{\mathrm{sc}}(U) \, \prod_X (1 + K(X; U)).$$

Theorem C.1 (KP stability)**.** *Under (6)–(7), Z is nonzero and $\log Z$ admits an absolutely convergent cluster expansion. Moreover, truncated correlations of local bounded observables decay exponentially with rate κ (up to constants).*

Proof. Write μ_{sc} itself as a polymer gas with activities $w(X)$. Then the combined model is a polymer gas whose activities are the union of w-polymers and K-polymers. The KP sum at a block B_0 is bounded by $\eta + \eta' < 1$. Hence the standard KP theorem for abstract polymer models applies: the cluster expansion converges absolutely, $Z \neq 0$, and the Ursell functions are exponentially summable. Exponential decay of truncated correlations follows from the usual bound of connected cluster weights by $e^{-\kappa \, \mathrm{dist}}$ when polymers have exponential weights. □

C.2 Deduction of Lemma 5.1

At the crossover scale k_* the effective density is of the form

$$\rho_{k_*}(U) = e^{-S^{\mathrm{loc}}_{k_*}(U)} \prod_X (1 + K_{k_*}(X;U)).$$

Expand the local part $e^{-S^{\mathrm{loc}}_{k_*}}$ in characters as in the standard strong-coupling expansion. Its induced polymer activities satisfy (6) when $\beta_{k_*} \le \beta_{\mathrm{sc}}$. The remainder K_{k_*} satisfies (7) provided $\|K_{k_*}\|$ is small enough. Then Theorem C.1 gives convergence and exponential decay, proving Lemma 5.1.

D Appendix D: Full Stroock–Zegarlinski sweeping proof of global LSI

This appendix replaces the sketches in Theorem 7.4 by a complete argument.

D.1 Setup

Let Λ be a finite index set (blocks/sites). For each $i \in \Lambda$ let $x_i \in G$. Let μ be a Gibbs measure on G^Λ with regular conditional probabilities $\mu_i^\eta(\,\mathrm{d}x_i)$. Let $E_i(f,f)$ be the local Dirichlet form with respect to the generator of the heat kernel on G acting on coordinate i:

$$E_i(f,f) := \int \|\nabla_i f\|^2 \,\mathrm{d}\mu,$$

where ∇_i is the Riemannian gradient on the ith copy of G.

Assume:

1. **Local LSI:** for all boundary conditions η, the conditional measure μ_i^η satisfies

$$\mathrm{Ent}_{\mu_i^\eta}(g^2) \le \frac{2}{\rho_{\mathrm{loc}}} \int \|\nabla g\|^2 \,\mathrm{d}\mu_i^\eta$$

   with $\rho_{\mathrm{loc}} > 0$ independent of η and i.

2. **Dobrushin contraction:** the coefficients c_{ij} defined in Definition 6 satisfy $q := \sup_i \sum_{j \ne i} c_{ij} < 1$.

D.2 Entropy decomposition

Let $(i_1, \ldots, i_{|\Lambda|})$ be an ordering of sites and let P_k be conditional expectation over the tail:

$$P_k f := \mathbb{E}_\mu[f \mid x_{i_k}, x_{i_{k+1}}, \ldots, x_{i_{|\Lambda|}}].$$

Then $P_{|\Lambda|+1} f = \mathbb{E}_\mu f$ and $P_1 f = f$. Set $f_k := P_k f$.

Using the chain rule for entropy,

$$\mathrm{Ent}_\mu(f^2) = \sum_{k=1}^{|\Lambda|} \mathbb{E}_\mu\Big[\mathrm{Ent}_{\mu^\eta_{i_k}}(f_k^2)\Big],$$

where η denotes the boundary variables $(x_{i_{k+1}}, \dots)$ on which f_k depends. Apply the local LSI to each conditional entropy:

$$\mathrm{Ent}_{\mu^\eta_{i_k}}(f_k^2) \le \frac{2}{\rho_{\mathrm{loc}}} \int \|\nabla_{i_k} f_k\|^2 \, \mathrm{d}\mu^\eta_{i_k}.$$

Therefore

$$\mathrm{Ent}_\mu(f^2) \le \frac{2}{\rho_{\mathrm{loc}}} \sum_{k=1}^{|\Lambda|} \mathbb{E}_\mu \|\nabla_{i_k} f_k\|^2. \tag{8}$$

D.3 Sweeping: controlling gradients under conditional expectation

The key lemma bounds $\nabla_i P_j f$ in terms of gradients of f.

Lemma D.1 (Gradient commutation / sweeping bound)**.** *For any $i \neq j$ and smooth f,*

$$\|\nabla_i P_j f\|_{L^2(\mu)} \le c_{ij} \, \|\nabla_j f\|_{L^2(\mu)} + \|\nabla_i f\|_{L^2(\mu)}.$$

Moreover, iterating along the ordering yields

$$\|\nabla_{i_k} f_k\|_{L^2(\mu)} \le \sum_{m \ge k} (C^{m-k}) \, \|\nabla_{i_m} f\|_{L^2(\mu)},$$

where C is the matrix (c_{ij}) restricted to the ordering and $\|C\|_{\ell^1 \to \ell^1} \le q$.

Proof. Fix $i \neq j$. Write $P_j f(x) = \int f(x_j, x_{\Lambda \setminus \{j\}}) \, \mu_j^{x_{\Lambda \setminus \{j\}}}(\,\mathrm{d}x_j)$. Differentiate in x_i:

$$\nabla_i P_j f = \int \nabla_i f \; \mathrm{d}\mu^\eta_j + \int f \, \nabla_i(\,\mathrm{d}\mu^\eta_j).$$

The first term is bounded by $\mathbb{E}[\|\nabla_i f\|^2]^{1/2}$ by Jensen. For the second term, use the dual Kantorovich representation of W_1 and the definition of c_{ij}: changes in x_i alter the conditional measure μ^η_j by at most c_{ji} in W_1, hence for any Lipschitz function φ,

$$\Big| \int \varphi \,(\,\mathrm{d}\mu^\eta_j - \,\mathrm{d}\mu^{\eta'}_j) \Big| \le c_{ji} d_G(x_i, x_i') \, \mathrm{Lip}(\varphi).$$

Apply this with $\varphi = f(\cdot)$ viewed as a function of x_j with Lipschitz constant controlled by $\|\nabla_j f\|$, and take an infinitesimal difference $x_i' \to x_i$ to obtain the bound by $c_{ij}\|\nabla_j f\|$. The iteration along the ordering is obtained by repeatedly applying the one-step inequality and summing a geometric series in q. □

D.4 Conclusion: global LSI

Insert Lemma D.1 into (8). Using $\sum_{n \ge 0} q^n = (1-q)^{-1}$ and Cauchy–Schwarz,

$$\sum_k \|\nabla_{i_k} f_k\|^2_{L^2} \le (1-q)^{-2} \sum_i \|\nabla_i f\|^2_{L^2}.$$

Therefore

$$\mathrm{Ent}_\mu(f^2) \le \frac{2}{\rho_{\mathrm{loc}}} (1-q)^{-2} \sum_i E_i(f, f),$$

i.e. μ satisfies a global LSI with constant $\rho \ge \rho_{\mathrm{loc}}(1-q)^2$, which is Theorem 7.4.

E Appendix E: Tightness details and identification of the OS continuum limit

This appendix expands the sketches in Section 8 and Section 8.4.

E.1 Uniform H^{-s} bounds from moment and mixing estimates

Fix a bounded domain $\Omega \subset \mathbb{R}^4$. Let $s > 2$ and let $\{e_n\}$ be an orthonormal basis of $H_0^s(\Omega)$. The H^{-s} norm of a distribution F supported in Ω is

$$\|F\|_{H^{-s}}^2 = \sum_{n\geq 1} |\langle F, e_n\rangle|^2.$$

Using Proposition 8.1 and Proposition 8.2, for each n we have $\sup_a \mathbb{E}|\langle F_a, e_n\rangle|^2 \leq C\|e_n\|_{H^s}^2 = C$. Moreover, the exponential mixing implies a uniform summability in n when weighted by $(1+\lambda_n)^{-s}$. Consequently

$$\sup_a \mathbb{E}\|F_a\|_{H^{-s}(\Omega)}^2 < \infty,$$

which is a quantitative tightness bound in the Hilbert space $H^{-s}(\Omega)$ (by Prokhorov in separable Hilbert spaces).

E.2 Mitoma implies tightness in $\mathcal{S}'$

For completeness we state the version of Mitoma used.

Theorem E.1 (Mitoma). *A family of $\mathcal{S}'(\mathbb{R}^d)$-valued random variables $\{X_\alpha\}$ is tight iff for every test function $\phi \in \mathcal{S}(\mathbb{R}^d)$ the family of real random variables $\{\langle X_\alpha, \phi\rangle\}$ is tight in $\mathbb{R}$.*

By the uniform second moment bounds proved above, $\{\langle F_a, \phi\rangle\}$ is tight for each ϕ, hence $\{F_a\}$ is tight.

E.3 Passing OS axioms to the limit

Let Θ be time reflection and let $\mathcal{A}_+$ be the algebra of observables supported in positive times. Reflection positivity on the lattice is the statement that for any $F = \sum_i c_i F_i$ with $F_i \in \mathcal{A}_+$,

$$\mathbb{E}_{\mu_a}[\Theta F \cdot F] = \sum_{i,j} \overline{c_i} c_j \, \mathbb{E}_{\mu_a}[\Theta F_i \cdot F_j] \geq 0.$$

If $S_a(F_i, F_j) \to S(F_i, F_j)$ along a subsequence, then by taking limits in this finite matrix inequality we obtain $\sum_{i,j} \overline{c_i} c_j S(\Theta F_i, F_j) \geq 0$, i.e. reflection positivity for the limit. The remaining OS axioms (Euclidean invariance, symmetry, clustering) are linear/inequality statements and pass similarly.

E.4 Mass gap in the continuum

Assume the lattice exponential clustering holds uniformly in physical units. For a local observable O with $\mathbb{E}[O] = 0$, define the two-point function $G_a(t) = \mathbb{E}_{\mu_a}[O(t)O(0)]$. Uniform clustering gives $|G_a(t)| \leq Ce^{-\Delta t}$. Passing to the limit yields the same bound for $G(t)$. By OS reconstruction, $G(t) = \langle \psi, e^{-tH}\psi\rangle$. If H had spectrum in $(0, \Delta)$ in this sector, $G(t)$ would decay strictly slower than $e^{-\Delta t}$. Thus the spectrum above the vacuum is bounded below by Δ.

This completes the tightness/identification gap.

F Appendix F: Polymer RG map, contraction, and plaquette moment bounds

This appendix supplies full details for the two remaining "Proof outline" steps that were still sketched in the main text: Lemma 4.13 (polymer contraction) and Proposition 8.1 (uniform plaquette moments). The arguments are standard once one works with a finite-range covariance decomposition and analytic polymer norms, but we record them explicitly.

F.1 Polymer algebra and the circle product

Let $\mathbb{B}_k$ be the set of k-blocks. A *polymer function* is a map F that assigns to each finite $X \subset \mathbb{B}_k$ a function $F(X;\cdot)$ of the field, with $F(\varnothing) = 1$. Given polymer functions F, G we define the *circle product* (convolution over disjoint unions) by

$$(F \circ G)(X) := \sum_{X = Y \sqcup Z} F(Y)\, G(Z),$$

where the sum is over all decompositions of X into disjoint (possibly empty) subsets. With this product, polymer functions form a commutative algebra.

If K is a polymer activity with $K(\varnothing) = 0$, define the polymer function $(1+K)$ by $(1+K)(\varnothing) = 1$ and $(1+K)(X) = K(X)$ for $X \neq \varnothing$. Then the formal product $\prod_X (1 + K(X))$ used in (4) is precisely the evaluation at the *full polymer* $X = \mathbb{B}_k$ of the exponential in the circle algebra:

$$\prod_{X \subset \mathbb{B}_k} (1 + K(X)) \quad \leftrightarrow \quad (1+K)^{\circ \mathbb{B}_k}$$

and absolute convergence is ensured by KP-type bounds (Lemma 4.12).

F.2 One-scale fluctuation integration as a polymer map

Fix a finite-range covariance piece $C_{k,j}$ from Lemma 4.5 (or Appendix A). Let $\mathbb{E}_j$ denote Gaussian expectation with covariance $C_{k,j}$ (in a small-field chart, on the gauge-fixed slice).

Write the scale-k density as

$$\rho_k = I_k \cdot (1 + K_k), \qquad I_k := e^{-S_k^{\mathrm{loc}}}.$$

Because S_k^{loc} is local, I_k factorizes over blocks (up to a fixed finite overlap that can be absorbed into a redefinition of blocks). For the contraction estimate it is enough to treat it as exactly block-factorized:

$$I_k(U) = \prod_{b \in \mathbb{B}_k} I_{k,b}(U), \qquad I_{k,b} \text{ depends only on links in } b^{\square}.$$

Define the single-layer RG transform

$$\mathcal{T}_j(I, K)(U) := \mathbb{E}_j\big[I(U + \zeta)\,(1+K)(U+\zeta)\big],$$

where U is the background (coarse) field and ζ is the fluctuation at that layer.

The output $\mathcal{T}_j(I,K)$ is again a polymer function, and by the BKAR connected expansion (Appendix B) it can be written as

$$\mathcal{T}_j(I,K)(U) = I'(U) \circ \big(1 + K'(U)\big), \tag{9}$$

where:

- I' is an updated local factor (block-factorized, obtained from the $X = \{b\}$ connected parts),
- K' is a polymer activity supported on connected polymers, obtained from $|X| \geq 2$ connected parts.

Moreover, the connected polymer bound Proposition B.2 implies

$$\|K'\|_{k,\kappa,r} \ \leq \ C\left(\|K\|_{k,\kappa,r} + \|V\|_{\mathrm{an}}\right)^2, \tag{10}$$

where V is the (small) non-quadratic part of the local interaction at that layer and $\|\cdot\|_{\mathrm{an}}$ is the analytic norm used in Appendix B. The square appears because $|X| \geq 2$ polymers require at least two vertices.

Iterating over all layers $j = 0, 1, \ldots$ at scale k and reblocking gives the full RG step.

F.3 Proof of Lemma 4.13

We now prove the contraction bound stated in Lemma 4.13.

Proof of Lemma 4.13. Fix a scale k and suppose $(S_k^{\mathrm{loc}}, K_k) \in \mathcal{D}_k$, so in particular:

1. on $\mathcal{S}_k$, S_k^{loc} is uniformly convex transverse to gauge orbits (Lemma 4.3),
2. the fluctuation covariance admits a finite-range decomposition (Lemma 4.5 / Appendix A),
3. large-field polymers satisfy KP (Lemma 4.12).

Step 1: small-field layer integration produces only small connected polymers. On $\mathcal{S}_k$ we write $S_k^{\mathrm{loc}} = Q_k + V_k$ where Q_k is the quadratic Maxwell form and V_k is the remainder. Lemma 4.1–4.2 imply

$$\|V_k\|_{\mathrm{an}} \lesssim \beta_k B_k^2$$

in the analytic norm controlling derivatives up to order r, because V_k starts at quartic order and $\|A\| \leq B_k$. Choose the small-field cutoff so that $\beta_k B_k^2 \leq \varepsilon_{\mathrm{sf}}$ for a universal $\varepsilon_{\mathrm{sf}}$. (Recall B_k is an RG-chosen cutoff; it can be made k-independent by taking it small enough.)

Apply the layer decomposition to the Gaussian measure for Q_k. At each layer, apply the BKAR connected expansion (Appendix B) to

$$\mathbb{E}_j\left[e^{-V_k(U+\zeta)}(1 + K_k)(U + \zeta)\right].$$

The connected polymers of size ≥ 2 are precisely the new layer activity K_j' in (9). Proposition B.2 gives an absolutely convergent expansion and yields

$$\|K_j'\|_{k,\kappa,r} \leq C\left(\|K_k\|_{k,\kappa,r} + \varepsilon_{\mathrm{sf}}\right)^2. \tag{11}$$

Step 2: scaling gives a linear contraction on the irrelevant sector. After completing all layers at scale k, we reblock by factor L and rescale fields to compare norms at scale $k+1$. For a polymer activity built from local monomials of engineering dimension $4 + \Delta$ ($\Delta > 0$), Lemma 4.9 gives a factor $L^{-\Delta}$ in the coefficient under rescaling. Because our polymer norm $\|\cdot\|_{k,\kappa,r}$ controls r derivatives and includes a regulator penalizing large polymers, this scaling manifests as

$$\|\mathcal{S}(K)\|_{k+1,\kappa,r} \leq L^{-\delta}\|K\|_{k,\kappa,r} \tag{12}$$

for $\delta = \min \Delta$ over the irrelevant sector (strictly positive in 4D once vacuum energy and Maxwell term are extracted).

Step 3: extraction removes marginal/relevant contributions and leaves only irrelevant terms. The RG-generated local part contains a vacuum term and a Maxwell term, plus higher-dimension operators. By definition of the extraction operator $\mathcal{P}$, we absorb the vacuum term and the Maxwell coefficient into S^{loc}_{k+1}. The remainder $(1-\mathcal{P})$ is irrelevant, hence subject to (12). This step is purely algebraic and does not require smallness.

Step 4: closing the inequality. Combine (11) for all layers and use that the number of layers per RG step is $O(1)$ (depending on how one chooses the finite-range decomposition). The contribution quadratic in $\|K_k\|$ is bounded by $C\|K_k\|^2$. For $\|K_k\| \le \varepsilon$ small enough we have $C\|K_k\|^2 \le \frac{1}{2}L^{-\delta}\|K_k\|$. The purely local interaction part produces a bounded irrelevant remainder of order $\varepsilon_{\mathrm{sf}}^2$; writing $\varepsilon_{\mathrm{sf}} \sim g_k^{p/2}$ gives the Cg_k^p term in Lemma 4.13. Finally, large-field contributions are controlled separately: KP implies they can be absorbed into a polymer activity whose norm is $O(e^{-c\beta_k B_k^2})$, which is dominated by Cg_k^p for large β_k and by strong-coupling KP once β_k is small.

Putting these together yields

$$\|K_{k+1}\|_{k+1,\kappa,r} \le L^{-\delta}\|K_k\|_{k,\kappa,r} + Cg_k^p,$$

as claimed. □

F.4 Proof of Proposition 8.1

Proof of Proposition 8.1. Fix a plaquette p. Split the expectation into small- and large-field events at the microscopic scale:

$$\mathbb{E}|Z_F(a)P(U_p)|^p = \mathbb{E}\big[|Z_F(a)P(U_p)|^p \mathbf{1}_{\mathcal{S}_0}\big] + \mathbb{E}\big[|Z_F(a)P(U_p)|^p \mathbf{1}_{\mathcal{L}_0}\big].$$

Small fields. On $\mathcal{S}_0$, in a block gauge chart we may write $U_p = \exp(F_p)$ with $\|F_p\| \lesssim B_0$ and $F_p = (dA)_p + O(A^2)$. Since P agrees with log near $\mathbf{1}$, $P(U_p) = F_p$ on $\mathcal{S}_0$. Use Lemma 4.2 to write F_p as a linear functional of the link variables plus a quadratic error:

$$F_p = L_p(A) + R_p(A), \qquad \|R_p(A)\| \le C \sum_{\ell \in \partial p} \|A_\ell\|^2.$$

The distribution of A under μ_a restricted to $\mathcal{S}_0$ is log-concave with Hessian $\succeq c\beta I$ (Lemma 4.3 plus gauge-fixing). Apply the Brascamp–Lieb moment bound (Lemma 4.4) to the linear functional $L_p(A)$ to obtain

$$\mathbb{E}_{\mathcal{S}_0}\|L_p(A)\|^p \le C_p\, \beta^{-p/2}.$$

For the remainder, $\|R_p(A)\| \le CB_0 \max_{\ell\in\partial p}\|A_\ell\|$ on $\mathcal{S}_0$, hence by the same moment bounds it is of smaller order. Thus

$$\mathbb{E}\big[|P(U_p)|^p \mathbf{1}_{\mathcal{S}_0}\big] \le C_p \beta^{-p/2}.$$

In physical units $U_\ell = \exp(gaA_{\mathrm{cont}})$, so $\beta^{-1/2} \sim g \sim Z_F(a)^{-1}$ and hence $|Z_F(a)P(U_p)| \sim a^2|F_{\mathrm{cont}}|$ on small fields. This yields the desired scaling $\le C_p a^{2p}$ after matching renormalization conventions.

Large fields. On $\mathcal{L}_0$, the variable $|P(U_p)|$ is bounded by a constant depending only on G (because P is bounded and G is compact), so

$$\mathbb{E}\big[|Z_F(a)P(U_p)|^p \mathbf{1}_{\mathcal{L}_0}\big] \le C\, Z_F(a)^p\, ¶(\mathcal{L}_0).$$

Large-field suppression (Lemma 4.11 and KP) gives $\P(\mathcal{L}_0) \leq \exp(-c\beta B_0^2)$ for fixed B_0. Along the asymptotically free trajectory $\beta \sim c\log(1/a)$, hence $\exp(-c\beta) \leq a^{c'}$ for some $c' > 0$. Choosing p fixed and a small yields a bound by $C_p a^{2p}$ after adjusting constants (or simply by noting $a^{c'} \leq a^{2p}$ for a small if $c' \geq 2p$, which can be arranged by taking the cutoff B_0 slightly larger).

Combining the small- and large-field estimates gives the stated uniform bound. $\square$